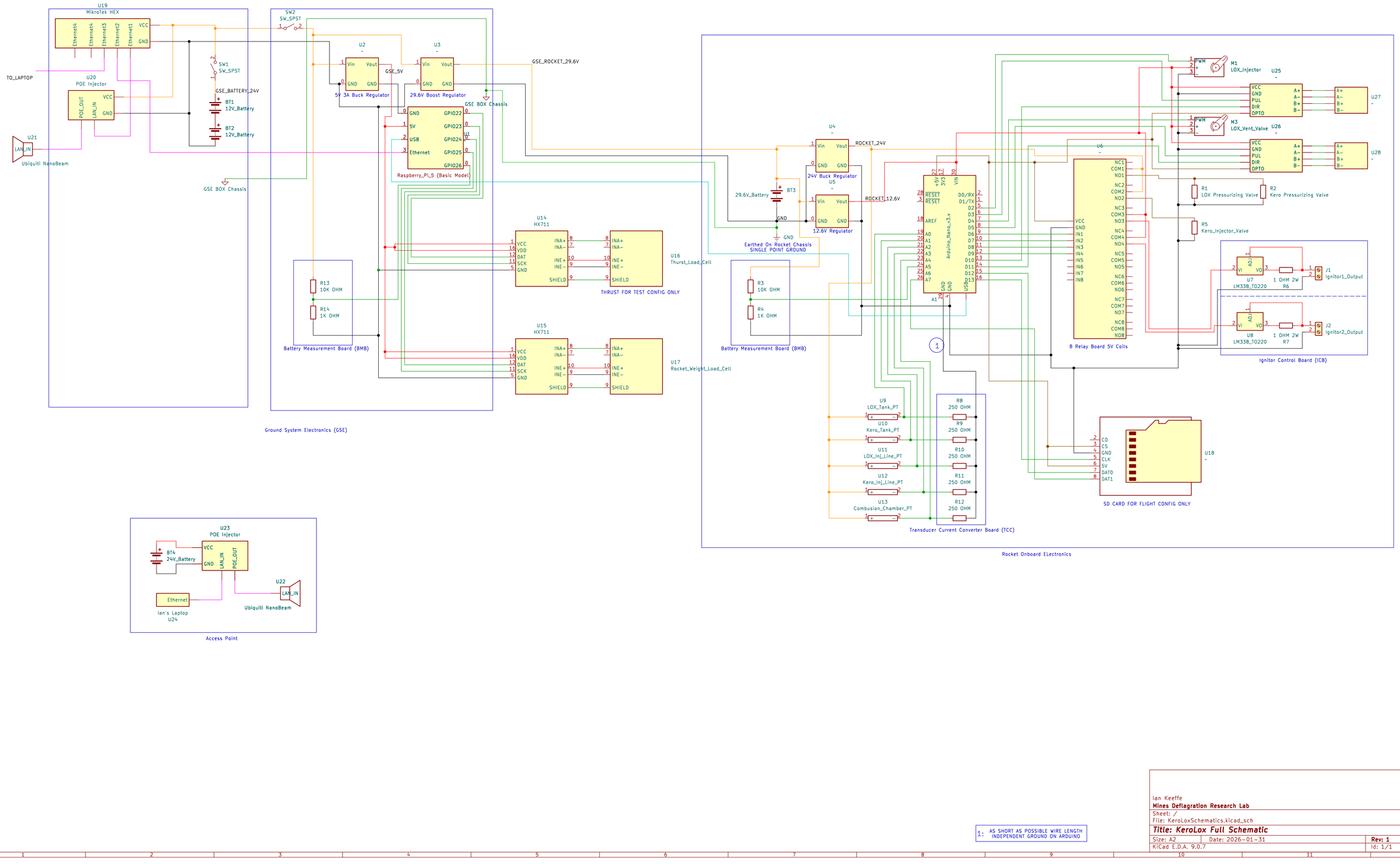

U19
MikroTek HEX
Ethernet4
Ethernet4
Ethernet3
Ethernet2
Ethernet1
VCC
GND
TO_LAPTOP
U20
POE Injector
POE_OUT
LAN_IN
VCC
GND
GSE_BATTERY_24V
SW1
SW_SPST
BT1
12V_Battery
BT2
12V_Battery
U21
LAN_IN
Ubiquiti NanoBeam
GSE BOX Chassis
SW2
SW_SPST
U2
Vin
Vout
GND
GND
5V 3A Buck Regulator
GSE_5V
U3
29.6V Boost Regulator
GSE_ROCKET_29.6V
GSE BOX Chassis
GND
GPIO22
5V
GPIO23
USB
GPIO24
Ethernet
GPIO25
GPIO26
Raspberry_Pi_5 (Basic Model)
U14
HX711
VCC
VDD
DAT
SCK
GND
INA+
INA−
INE+
INE−
SHIELD
U16
Thurst_Load_Cell
THRUST FOR TEST CONFIG ONLY
R13
10K OHM
R14
1K OHM
Battery Measurement Board (BMB)
U15
HX711
U17
Rocket_Weight_Load_Cell
Ground System Electronics (GSE)
U4
Vin
Vout
GND
GND
24V Buck Regulator
ROCKET_24V
U5
ROCKET_12.6V
12.6V Regulator
29.6V_Battery
BT3
GND
Earthed On Rocket Chassis
SINGLE POINT GROUND
R3
10K OHM
R4
1K OHM
Battery Measurement Board (BMB)
RESET
AREF
A0
A1
A2
A3
A4
A5
A6
A7
+5V
3V3
VIN
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
USB
Arduino_Nano_v3.x
U6
NC1
COM1
NO1
NC2
COM2
NO2
NC3
COM3
NO3
NC4
COM4
NO4
NC5
COM5
NO5
NC6
COM6
NO6
NC7
COM7
NO7
NC8
COM8
NO8
VCC
GND
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
8 Relay Board 5V Coils
PWM
M1
LOX_Injector
M3
LOX_Vent_Valve
U25
U26
VCC
GND
PUL
DIR
OPTO
A+
A−
B+
B−
U27
U28
R1
LOX Pressurizing Valve
R2
Kero Pressurizing Valve
R5
Kero_Injector_Valve
U7
LM338_TO220
1 OHM 2W
R6
J1
Ignitor1_Output
U8
LM338_TO220
1 OHM 2W
R7
J2
Ignitor2_Output
Ignitor Control Board (ICB)
U9
LOX_Tank_PT
U10
Kero_Tank_PT
U11
LOX_Inj_Line_PT
U12
Kero_Inj_Line_PT
U13
Combusion_Chamber_PT
R8
250 OHM
R9
250 OHM
R10
250 OHM
R11
250 OHM
R12
250 OHM
Transducer Current Converter Board (TCC)
CD
CS
GND
CLK
5V
DAT0
DAT1
U18
SD CARD FOR FLIGHT CONFIG ONLY
Rocket Onboard ELectronics
U23
POE Injector
BT4
24V_Battery
VCC
GND
LAN_IN
POE_OUT
U22
LAN_IN
Ethernet
Ian's Laptop
U24
Ubiquiti NanoBeam
Access Point
1:
AS SHORT AS POSSIBLE WIRE LENGTH
INDEPENDENT GROUND ON ARDUINO
Ian Keeffe
Mines Deflagration Research Lab
Sheet: /
File: KeroLoxSchematics.kicad_sch
Title: KeroLox Full Schematic
Size: A2
Date: 2026-01-31
Rev: 1
KiCad E.D.A. 9.0.7
Id: 1/1